

AP EAMCET-2017-Engineering-26 Apr-Shift-1
Questions Physics

81. In the determination of the internal resistance of a cell with a potentiometer, the error in the measurement of the balancing length is ± 1 mm. When the cell alone is connected in the circuit, the balancing length is obtained at 60 cm and when the cell is shunted with a resistance of $10\ \Omega \pm 2\%$, the balancing length is obtained at 50 cm. The error in the determination of the internal resistance of the cell is _____
- (1) 1.8%
 - (2) 2.4%
 - (3) 4.2%
 - (4) 5.6%
82. A bird is tossing (flying to and fro) between two cars moving towards each other on a straight road. One car has speed of 54 kmh^{-1} while the other has the speed of 36 kmh^{-1} . The bird starts moving from first car towards the other and is moving with the speed of 36 kmh^{-1} where the two cars were separated by 36 km. The total distance covered by the bird before the cars meet each other is

- (1) 1440 m
- (2) 24400 m
- (3) 244 m
- (4) 14400 m

83. A body is projected from the top of a tower with a velocity $\vec{u} = 3\hat{i} + 4\hat{j} + 5\hat{k} \text{ ms}^{-1}$ where $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors along east, north and vertically upwards respectively. If the height of the tower is 30 m, horizontal range of the body on the ground is

($g = 10 \text{ ms}^{-2}$)

- (1) 12 m
- (2) 9 m
- (3) 25 m
- (4) 15 m

84. Equation of a projectile is given by $y = Px - Qx^2$ where P , Q are constants. The ratio of maximum height to range of the projectile is

- (1) $\frac{P^2}{Q}$
- (2) $4P$

(3) $\frac{P}{4}$

(4) $\frac{Q^2}{2P}$

85. A person of 60 kg mass is in a lift which is coming down such that the man exerts a force of 150 N on the floor of the lift. Then the acceleration of the lift is ($g = 10 \text{ ms}^{-2}$)

(1) 40.0 ms^{-2}

(2) 15.0 ms^{-2}

(3) 22.5 ms^{-2}

(4) 7.5 ms^{-2}

86. The situations are shown in fig (a) & (b), in each case, $m_1 = 3 \text{ kg}$ and $m_2 = 4 \text{ kg}$. If a_1, a_2 are the respective accelerations of the blocks in these situation, then the values of a_1 and a_2 are respectively $[g = 10 \text{ ms}^{-2}]$

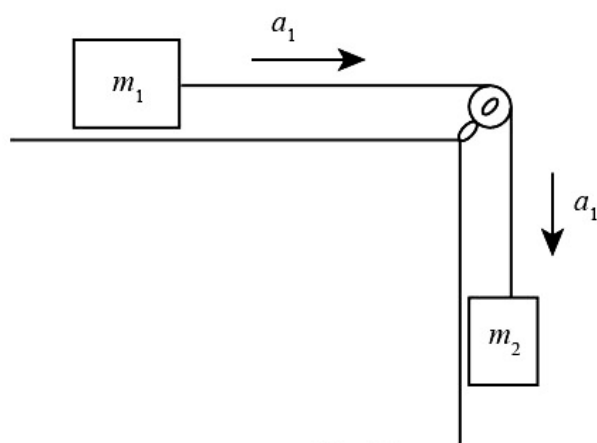


Fig. (a)

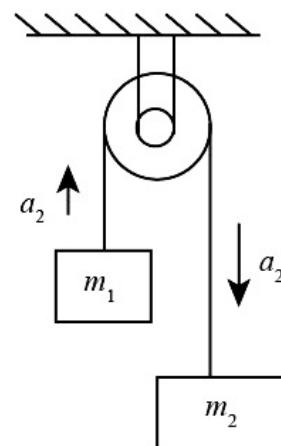


Fig. (b)

(1) $\frac{10}{7} \text{ ms}^{-2}, \frac{25}{7} \text{ ms}^{-2}$

(2) $\frac{30}{7} \text{ ms}^{-2}, \frac{5}{7} \text{ ms}^{-2}$

(3) $\frac{40}{7} \text{ ms}^{-2}, \frac{10}{7} \text{ ms}^{-2}$

(4) $\frac{20}{7} \text{ ms}^{-2}, \frac{10}{7} \text{ ms}^{-2}$

87. A girl of mass 50 kg swinging on a cradle. If she moves with a velocity of 2 ms^{-1} upward in a direction making an angle 60° with the vertical, then the power generated is ($g = 9.8 \text{ ms}^{-2}$)

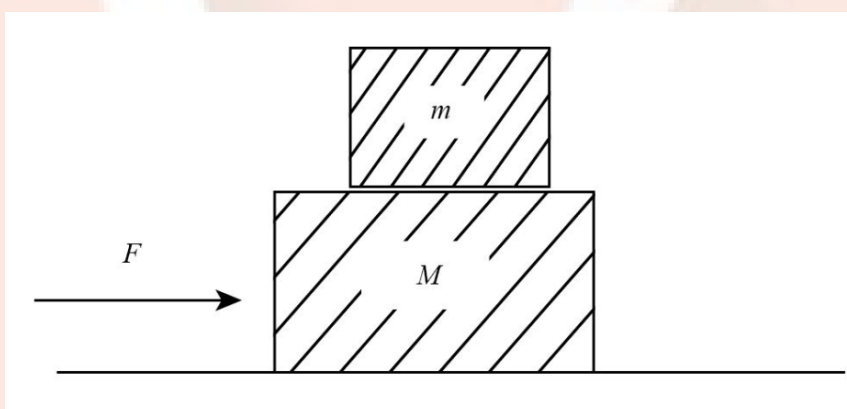
(1) 980 W

(2) $490\sqrt{2} \text{ W}$

(3) $490\sqrt{3}$ W

(4) 245 W

88. Two blocks of masses ' M ' and ' m ' are placed on one another on a smooth horizontal surface as shown in the figure. The force ' F ' is acting on the mass ' M ' horizontally during time interval ' t '. Assuming no relative sliding between the blocks, the work done by friction on the blocks is _____



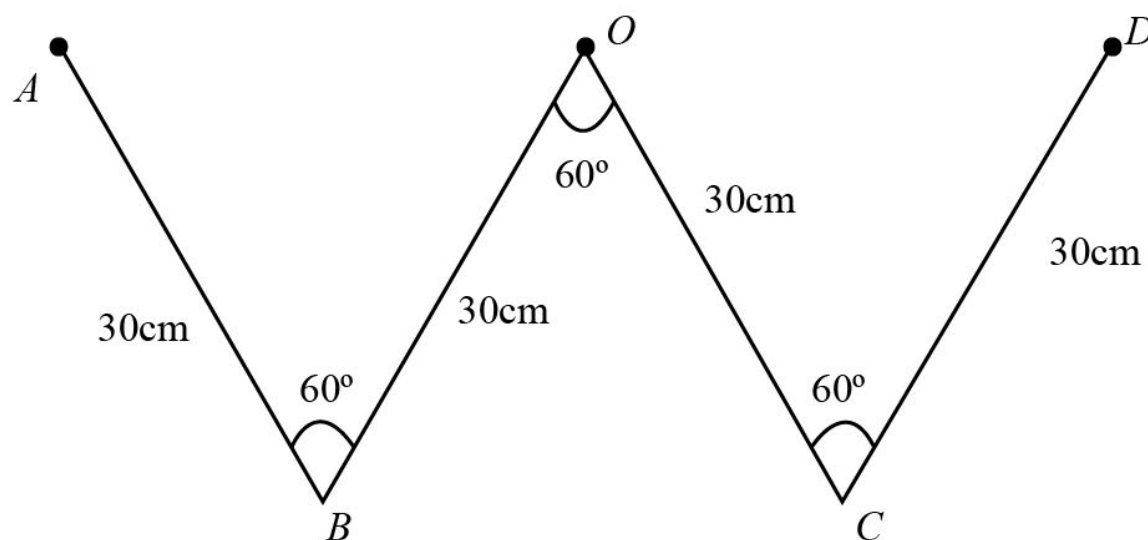
(1) $\frac{mF^2t^2}{2(M+m)^2}$

(2) $\frac{F^2t^2}{(M+m)}$

(3) $\frac{M+m}{mt^2}$

(4) $\frac{Ft}{2(M+m)}$

89. A uniform thin rod of 120 cm length 1600 g mass is bent as shown in the figure. The moment of inertia of the bent rod about an axis passing through the point 'O' and perpendicular to the plane of the paper is _____ kg m^2 .



- (1) 0.120
- (2) 0.084
- (3) 0.360
- (4) 0.018

90. A solid sphere is projected up along an inclined plane of inclination 30° with the horizontal with a speed of 4 ms^{-1} . If it rolls without slipping, the maximum distance traversed by it is _____ ($g = 10 \text{ ms}^{-2}$)

- (1) 1.12 m

(2) 22.4 m

(3) 112 m

(4) 2.24 m

91. Time period of a simple pendulum of length ' L ' is T_1 . Time period of a uniform rod of same length ' L ' suspended from one end and oscillating in a vertical plane is T_2 . Amplitude of oscillation is small in both the cases. Then $\frac{T_1}{T_2}$ is

(1) $\sqrt{\frac{4}{3}}$

(2) 1

(3) $\sqrt{\frac{3}{2}}$

(4) $\sqrt{\frac{2}{3}}$

92. Two bodies of masses 4 m and 9 m are separated by a distance ' r '. The gravitational potential at a point on this line joining them where the gravitational field becomes zero is

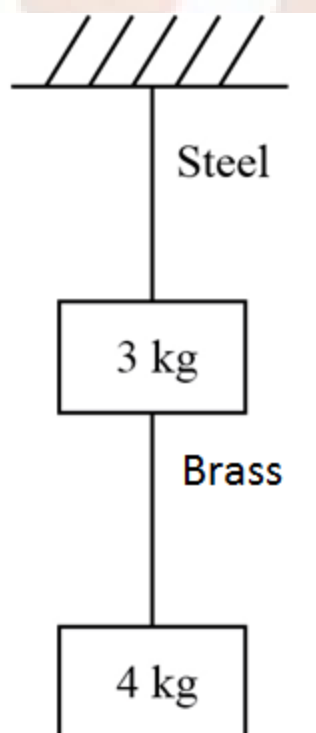
(1) $\frac{-4Gm}{r}$

$$(2) \frac{-25Gm}{r}$$

$$(3) \frac{-9Gm}{r}$$

$$(4) \frac{-13Gm}{r}$$

93. The ratios of lengths, areas of cross-section and Young's moduli of steel to that of brass wires shown in the figure are a , b and c respectively. The ratio of increase in the lengths of brass to that of steel wires is [Assume that the masses of steel and Brass wires are negligible]



(1) $\frac{4a}{7bc}$

(2) $\frac{7bc}{4a}$

(3) $\frac{4ab}{7a}$

(4) $\frac{7a}{4bc}$

94. Fully filled open water tank has two holes on either side of its walls. One is a square hole of side x cm at a depth of 2 m from the top, and the other hole is equilateral triangle of side 4 cm at a depth of 6 m from the top. If the rate of flow of water is same from both the holes, then 'x' is

(1) 12 cm

(2) 3.46 cm

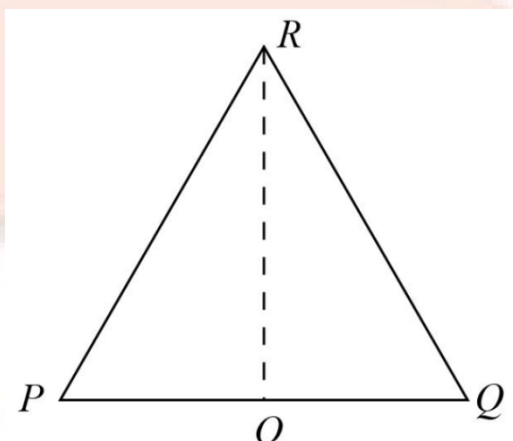
(3) 6.92 cm

(4) 1.73 cm

95. Three uniform thin aluminum rods each of length 2 m form an equilateral triangle PQR as shown in the figure. The mid point of the rod PQ is at the origin of the coordinate system. If the temperature of the system of rods increases by 50°C , the increase

in y -coordinate of the centre of mass of the system of the rods is _____ mm.

(Coefficient of volume expansion of aluminium $= 12\sqrt{3} \times 10^{-6} \text{ K}^{-1}$)



- (1) 0.05
- (2) 0.1
- (3) 0.2
- (4) 0.8

96. A wall is made of equally thick layers 'P' and 'Q' of different materials. Thermal conductivity of 'Q' is half of that of the 'P'. In the steady state, if the temperature difference across the wall is 24°C , then the temperature difference across the layer 'P' is _____

- (1) 12°C
- (2) 8°C

(3) 16°C

(4) 4°C

97. One mole of a gas expands such that its volume ' V ' changes with absolute temperature ' T ' in accordance with the relation $V = KT^2$ where ' K ' is a constant. If the temperature of the gas changes by 60°C , then work done by the gas is (R is universal gas constant).

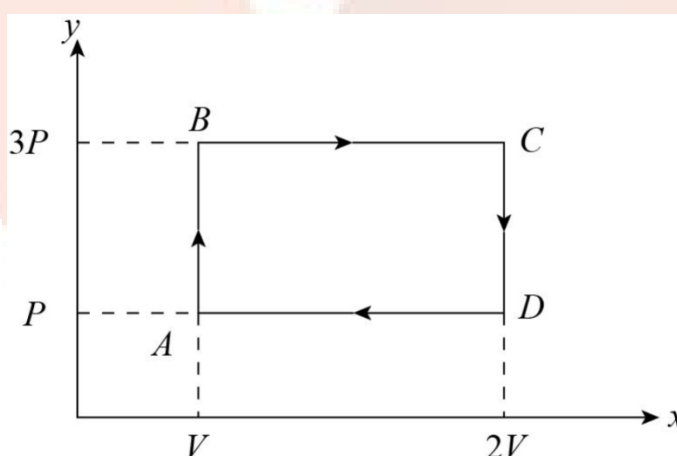
(1) $120R$

(2) $R\ln 60$

(3) $KR\ln 60$

(4) $40KR$

98. A monatomic ideal gas goes through a cyclic process as shown in the figure. The efficiency of this process is



(1) 78%

(2) 62%

(3) 42%

(4) 21%

99. If the average translational kinetic energy of a molecule in a gas is equal to the kinetic energy of an electron accelerating from rest through 10 volt, then the temperature of the gas molecule is (Boltzmann constant = $1.38 \times 10^{-23} \text{ JK}^{-1}$)

(1) 73.7 K

(2) $77.3 \times 10^3 \text{ K}$

(3) $7.73 \times 10^3 \text{ K}$

(4) 730 K

100. The transverse displacement of a string linear density 0.01 kgm^{-1} , clamped at its ends is given by

$$Y_{(x,t)} = 0.03 \sin\left(\frac{2\pi x}{3}\right) \cos(60\pi t) \text{ where } x \text{ and } y \text{ are in meters}$$

and time ' t ' is in seconds. Tension in the string is

(1) 81 N

(2) 162 N

(3) 36 N

(4) 9 N

101. A closed organ pipe of length 'L' and an open organ pipe contain gases of densities ρ_1 and ρ_2 respectively. The compressibility of gases is equal in both the pipes. If the frequencies of their first overtones are same, then the length of the open organ pipe is

(1) $\frac{L}{3}$

(2) $\frac{4L}{3}$

(3) $\frac{4L}{3} \sqrt{\frac{\rho_1}{\rho_2}}$

(4) $\frac{4L}{3} \sqrt{\frac{\rho_2}{\rho_1}}$

102. Two point sources S_1 and S_2 are 24 cm apart. Where should a convex lens of focal length 9 cm be placed in between them so that the images of both sources are formed at the same place?

(1) 8 cm

(2) 6 cm

(3) 10 cm

(4) 12 cm

103. Two slits separated by 0.5 mm are illuminated by light of wavelength 500 nm. The screen is at a distance of 120 cm from the slits. The phase difference between the interfering waves at a point 3 mm on the screen from the central bright fringe is _____

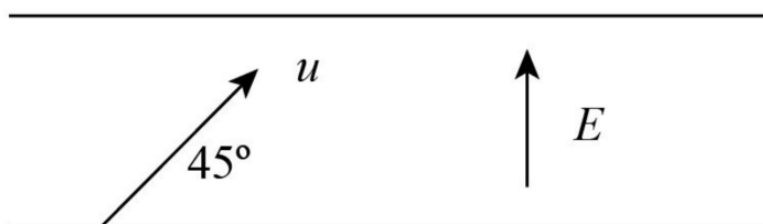
(1) π

(2) 3π

(3) 5π

(4) 7π

104. The uniform electric field intensity between the two plates of a parallel plate capacitor is $1 \times 10^3 \text{ Vm}^{-3}$ acting vertically upwards as shown in the figure. The plates are sufficiently long and have separation 2 cm. A particle of negative charge $1 \mu\text{C}$ and mass 2 g is projected at an angle 45° with the electric field from the lower plate with a velocity ' u '. The maximum velocity acquired by the particle, if it is not hit the upper plate is



(1) 0.1 ms^{-1}

(2) 0.2 ms^{-1}

(3) 2 ms^{-1}

(4) 1 ms^{-1}

105. An infinitely long thin straight wire has uniform linear charge density of $\frac{1}{3} \text{ Cm}^{-1}$. Then the magnitude of the force acting on a charge $3 \text{ } \mu\text{C}$ situated at a point of 18 cm away from the wire is

$$\left(\frac{1}{4\pi \epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-2} \right)$$

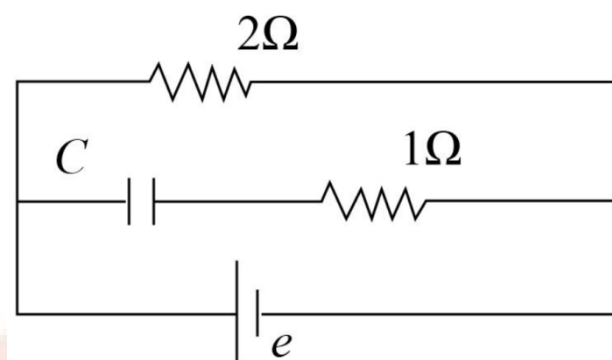
(1) 10^5 N

(2) $2 \times 10^5 \text{ N}$

(3) $\frac{1}{3} \times 10^6 \text{ N}$

(4) $3 \times 10^{11} \text{ N}$

106. In steady state, a capacitor of capacitance $2 \text{ } \mu\text{F}$ is charged to $4 \text{ } \mu\text{C}$, as shown in figure. If internal resistance of the cell is $0.5 \text{ } \Omega$, the emf of the Cell is



- (1) 4 V
- (2) 2.5 V
- (3) 5 V
- (4) 2 V

107. An electrostatic paint sprayer has a metal sphere of diameter 18 cm and at a potential of 25 kV. If it sprays paint droplets, the charge on the metal sphere is

- (1) $0.5 \mu\text{C}$
- (2) $0.25 \mu\text{C}$
- (3) $2.5 \mu\text{C}$
- (4) $25 \mu\text{C}$

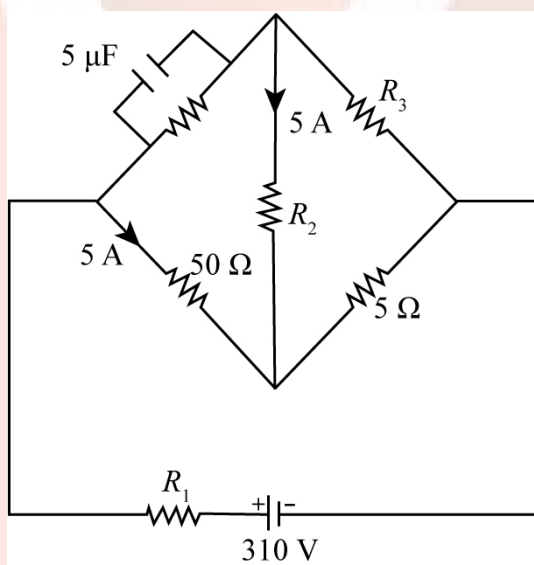
108. When the terminals of a cell are connected by a wire of resistance 4Ω , the potential difference across the cell is 1.6 V. If a wire of the same resistance is connected in parallel with the

first, the potential difference becomes 1.33 V. The emf and internal resistance of the cell are respectively

- (1) 2 V, 2 Ω
- (2) 1 V, 2 Ω
- (3) 1 V, 1 Ω
- (4) 2 V, 1 Ω

109. If the charge on the capacitor is 1 mC in the given circuit, then

$$\frac{R_1 R_2}{R_3} = \text{_____ } \Omega$$



- (1) 0.4
- (2) 10
- (3) 6
- (4) 0.6

110. Two concentric coils of 20 turns each are placed in same plane. Their radii are 30 cm and 60 cm and carry 0.4 A and 0.6 A current respectively in opposite directions. The magnetic induction at the centre in Tesla is _____

(1) $\frac{5}{3}\mu_0$

(2) $\frac{8}{3}\mu_0$

(3) $\frac{2}{3}\mu_0$

(4) $\frac{10}{3}\mu_0$

111. Two long parallel conducting wires carrying currents are separated by a distance 'x' Work done per unit length in changing the distance between the wires is proportional to

(1) x

(2) $\frac{1}{x}$

(3) $\log_e x$

(4) $\frac{1}{\log_e x}$

112. Match the following

List - I

(a) High Retentivity

(b) High resistivity

(c) Low coercivity

(d) Negative susceptibility

List - II

(i) Telephone diaphragm

(ii) Diamagnet

(iii) To decrease eddy currents losses

(iv) Permanent magnet

The correct answer is

(1) a-i, b-ii, c-iii, d-iv

(2) a-iv, b-iii, c-i, d-ii

(3) a-i, b-iv, c-iii, d-ii

(4) a-iv, b-ii, c-i, d-iii

113. Assertion (A) : It is more difficult to push magnet into a coil with more number of turns.

Reason (R) : The emf induced in a coil opposes the motion of a magnet when it is moved towards the coil.

(1) Both A and R are true. R is a correct explanation of A.

(2) Both A and R are true. R is not correct explanation of A

(3) A is true, R is false

(4) A is false, R is true

114. When a coil is connected to AC supply of frequency 50 Hz, a current of 4 A flows in it and it consumes 240 W power. If the potential difference across the coil is 100 V, then the inductance value of the coil is

(1) $L = \frac{1}{5\pi} \text{ H}$

(2) $L = \frac{\pi}{5} \text{ H}$

(3) $L = (5\pi) \text{ H}$

(4) $L = \frac{1}{25\pi} \text{ H}$

115. The magnitude of electric field in an electromagnetic wave is 60 Vm^{-1} . Then the amplitude of magnetic field is

(1) $2 \times 10^7 \text{ T}$

(2) $6 \times 10^7 \text{ T}$

(3) $6 \times 10^{-7} \text{ T}$

(4) $2 \times 10^{-7} \text{ T}$

116. Both an electron and a photon have same de-Broglie wavelength of $1.2 \text{ \AA}$. The ratio of their energies is nearly

- (1) 1 : 1
- (2) 1 : 100
- (3) 1 : 100
- (4) 1 : 1000

117. The radius of orbit of an electron and the speed of electron in the ground state of hydrogen atom are 5.5×10^{-11} m and 4×10^6 m/s respectively. Then, the orbital period of this electron in the first excited state will be _____

- (1) 9.608×10^{-16} s
- (2) 8.9068×10^{-16} s
- (3) 7.806×10^{-16} s
- (4) 6.908×10^{-16} s

118. The half life of a stream of radioactive particles moving along a straight path with a constant kinetic energy of 4 eV is 1 minute. The percentage of particles which decay before travelling a distance of 3.6 km is _____

(Mass of the radioactive particles = 3.2×10^{-21} kg and charge of the electron = 1.6×10^{-19} C).

- (1) 37.5
- (2) 87.5
- (3) 75
- (4) 175

119. A Zener diode voltage regulator operated in the range 120–180 V produces a constant supply of 110 V and 250 mA to the load. If the maximum current is equally shared between the load and the Zener diode, then the values of load resistance (R_L) and series resistance (R_S) are respectively

- (1) $R_L = 70 \Omega$, $R_S = 280 \Omega$
- (2) $R_L = 440 \Omega$, $R_S = 140 \Omega$
- (3) $R_L = 440 \Omega$, $R_S = 1400 \Omega$
- (4) $R_L = 280 \Omega$, $R_S = 70 \Omega$

120. The process of recovering the modulating signal from the modulated carrier wave is called

- (1) Amplification
- (2) Rectification
- (3) Noise
- (4) Detection

AP EAMCET-2017-Engineering-26 Apr-Shift-1 Solutions – Physics

81. The formula for the internal resistance of the cell is,

$$r = R \left(\frac{l_1}{l_2} - 1 \right)$$

The percentage change in the internal resistance is,

$$\begin{aligned}\frac{\Delta r}{r} &= \frac{\Delta R}{R} \times 100 + \frac{\Delta \left(\frac{l_1}{l_2} \right)}{\left(\frac{l_1}{l_2} \right)} \times 100 \\ &= 4 + 0.2 \\ &= 4.2\%\end{aligned}$$

82. The total time required by the two cars to meet,

$$t = \frac{36}{54 + 36}$$

$$t = 0.4 \text{ h}$$

The distance covered by the bird before the cars meet each other is,

$$\begin{aligned}d &= 36 \times 0.4 \\ &= 14.4 \text{ km or} \\ &= 14400 \text{ m}\end{aligned}$$

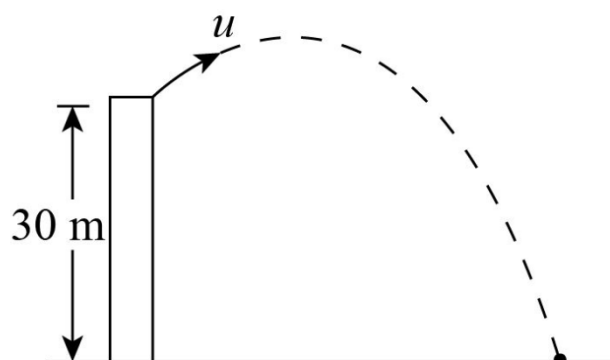
83. Given Data:

$$u = 5 \text{ m/s}$$

$$a = -10 \text{ m/s}^2$$

$$h = -30 \text{ m}$$

The figure below represents the projectile motion.



$$h = ut + \frac{1}{2}at^2$$

$$-30 = 5t - \frac{1}{2} \times 10 \times t^2$$

By solving the above equation.

$$t = 3 \text{ or } t = -2$$

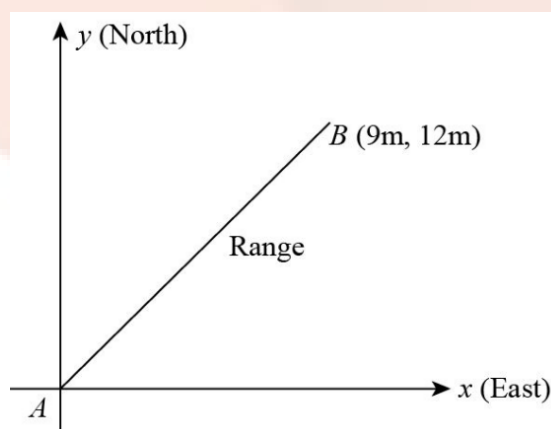
$$t = 3 \text{ s}$$

The distance covered in east direction is,

$$d_x = 3 \times 3 = 9 \text{ m}$$

The distance covered in north direction is,

$$d_y = 4 \times 3 = 12 \text{ m}$$



The range is,

$$R = \sqrt{9^2 + 12^2}$$

$$R = \sqrt{225} \text{ m}$$

$$R = 15 \text{ m}$$

84. The equation of the projectile is given as,

$$y = Px - Qx^2 \quad \dots\dots (I)$$

The distance travelled,

$$y = x \tan \theta - \frac{1}{2} g \left(\frac{x^2}{u^2 \cos^2 \theta} \right) \quad \dots\dots (II)$$

From equation (I) and (II),

$$P = \tan \theta$$

$$Q = \frac{g}{2u^2 \cos^2 \theta}$$

The formula for maximum height and range is,

$$H = \frac{u^2 \sin^2 \theta}{2g}$$

$$R = \frac{u^2 \sin 2\theta}{2g}$$

By dividing the two equations,

$$\frac{H}{R} = \frac{\sin^2 \theta}{2 \times \sin 2\theta}$$

$$\frac{H}{R} = \frac{\sin^2 \theta}{2 \times 2 \sin \theta \cos \theta}$$

$$\frac{H}{R} = \frac{\tan \theta}{4}$$

$$\frac{H}{R} = \frac{P}{4}$$

85. The formula for the resultant acceleration when the lift goes down,

$$a' = (g - a)$$

The force exerted by the man on the floor of the lift is,

$$F = ma'$$

$$F = m(g - a)$$

$$150 = 60(10 - a)$$

$$a = 7.5 \text{ m/s}^2$$

86. Given Data:

$$m_1 = 3 \text{ kg}, m_2 = 4 \text{ kg}$$

Consider the diagram as shown below,

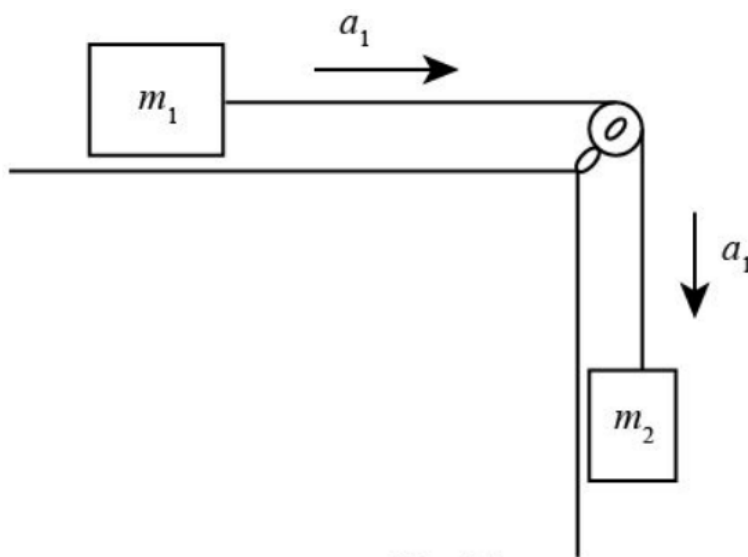


Fig. (a)

Balancing the force,

$$(m_1 + m_2)a_1 = m_2g$$

$$a_1 = \frac{m_2g}{m_1 + m_2}$$

$$a_1 = \frac{4 \times 10}{3 + 4} = \frac{40}{7} \text{ m/s}^2$$

Consider the diagram as shown below,

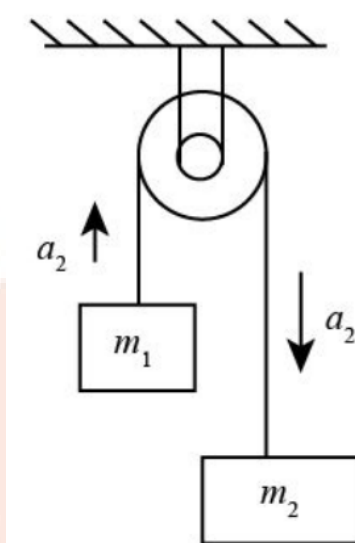


Fig. (b)

By balancing the forces,

$$(m_1 + m_2)a = (m_2 - m_1)g$$

$$a = \frac{m_2 - m_1}{(m_1 + m_2)}g$$

$$a = \frac{4 - 3}{4 + 3} \times 10$$

$$a = \frac{10}{7} \text{ m/s}^2$$

87. The work done to displace the girl can be calculated as,

$$W = mg \sin 60^\circ ds$$

$$W = mg \sin 60^\circ (v dt)$$

The power generated at that instant is,

$$P = mg \sin 60^\circ (v)$$

$$P = 50(9.8) \left(\frac{\sqrt{3}}{2} \right) (2)$$

$$P = 490\sqrt{3} \text{ W}$$

88. The acceleration of the block can be calculated as,

$$F = (m + M)a$$

$$a = \frac{F}{m + M}$$

Since there is no relative sliding, the frictional force on upper block is,

$$f = ma$$

$$f = m \left(\frac{F}{M + m} \right)$$

The distance covered can be calculated as,

$$D = \frac{1}{2}at^2$$

$$D = \frac{1}{2} \left(\frac{F}{M + m} \right) t^2$$

The work done by the frictional force is,

$$W = f \times s$$

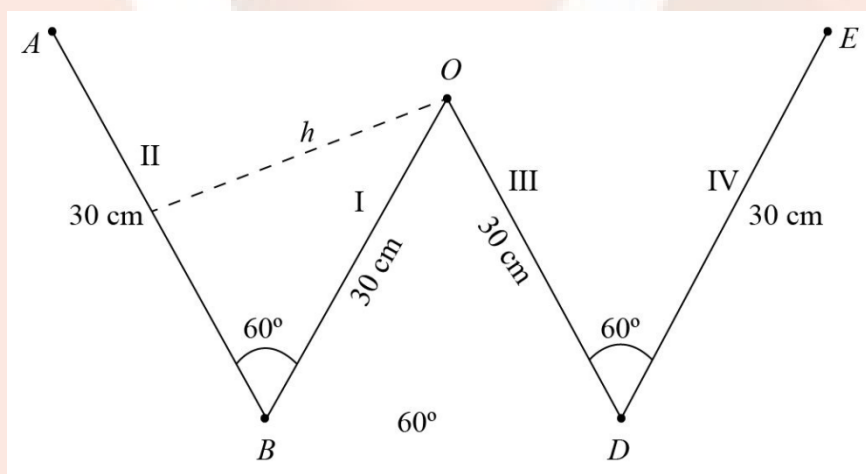
$$= \frac{mF}{M+m} \times \frac{1}{2} \frac{Ft^2}{m+m}$$

$$= \frac{mF^2t^2}{2(m+M)}$$

89. The mass of each rod is, $m = 400 \text{ g} = 0.4 \text{ kg}$

The length of each rod is, $l = 30 \text{ cm} = 0.3 \text{ m}$

The figure below represents the uniform rods,



$$h^2 = l^2 - \left(\frac{l}{2}\right)^2$$

$$h^2 = \frac{3}{4}l^2$$

The moment of inertia of I & III is same and II and IV is same.

The moment of inertia of the system,

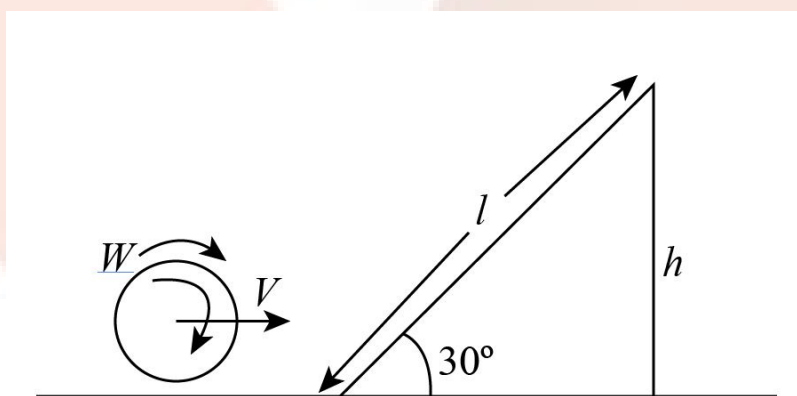
$$\begin{aligned}
 MI &= 2 \times (MI \text{ of } I) + 2 \times (MI \text{ of } II) \\
 &= \frac{ml^2}{3} \times 2 + 2 \left(\frac{ml^2}{12} + mh^2 \right) \\
 &= \frac{2}{3}ml^2 + 2ml^2 \left(\frac{10}{12} \right) \\
 &= \frac{7}{3}ml^2
 \end{aligned}$$

Substitute the values in the above equation.

$$\begin{aligned}
 MI &= \frac{7}{3} \times 0.4 \times 0.3 \times 0.3 \\
 &= 0.084 \text{ kg} \cdot \text{m}^2
 \end{aligned}$$

90. The velocity of the solid sphere is $v = 4 \text{ m/s}$

The figure below represents the solid sphere moves along the inclined plane.



The kinetic energy at the bottom, is equal to the sum of kinetic energy due to translation and rotation,

$$\begin{aligned} KE &= KE_t + KE_r \\ &= \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \\ &= \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{2}{5}mr^2\right)\left(\frac{v^2}{r^2}\right)^2 \\ &= \frac{1}{2}\left(\frac{7}{5}mv^2\right) \end{aligned}$$

From the conservation of energy, at height h ,

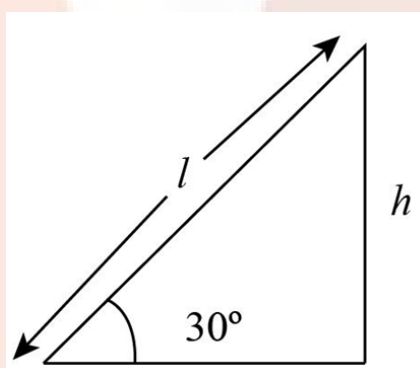
$$KE = mgh$$

$$\frac{1}{2}\left(\frac{7}{5}mv^2\right) = mgh$$

$$\frac{1}{2}\left(\frac{7}{5}(4)^2\right) = (10)h$$

$$h = 1.12 \text{ m}$$

The maximum distance travelled is,



$$l = \frac{h}{\sin 30^\circ}$$

$$l = \frac{1.12}{\sin 30^\circ}$$

$$l = 22.4 \text{ m}$$

91. The formula for the time period of the pendulum,

$$T = 2\pi \sqrt{\frac{L}{g}}$$

$$T \propto \sqrt{\frac{L}{g}}$$

The ratio of T_1 / T_2 is

$$\frac{T_1}{T_2} = \frac{\sqrt{L/g}}{\sqrt{2L/3g}}$$

$$\frac{T_1}{T_2} = \sqrt{\frac{3}{2}}$$

92. The position where the gravitation field is zero.

$$\frac{G(4m)}{x} = \frac{G(9m)}{(r-x)}$$

Here, x is distance is from 4 m mass.

$$\frac{4}{x} = \frac{9}{(r-x)}$$

$$4(r-x) = 9x$$

$$4r = 13x$$

$$x = \frac{4}{13}r$$

The point P is at a distance $\frac{4}{13}r$ from mass $4m$ and

$$\left(h - \frac{4}{13}r\right) = \frac{9r}{13} \text{ from mass } m.$$

$$v = -\frac{G(4m)}{\left(\frac{4}{13}r\right)} - \frac{G(9m)}{\frac{9r}{13}}$$

$$v = -\frac{26Gm}{r}$$

93. Given:

$$\text{Length: } \frac{L_s}{L_b} = a$$

$$\text{Area of cross section: } \frac{A_s}{A_b} = b$$

$$\text{Young's modules: } \frac{E_s}{E_b} = b$$

The formula for the increase in the length,

$$\frac{\Delta L}{L} = \frac{\sigma}{E}$$

$$\Delta L = \frac{F / A}{E} L$$

$$\Delta L = \frac{FL}{AE}$$

The ratio of change in length of steel and brass,

$$\frac{\Delta L_s}{\Delta L_b} = \left(\frac{F_s L_s}{A_s E_s} \right) \left(\frac{A_s E_s}{F_b L_b} \right)$$

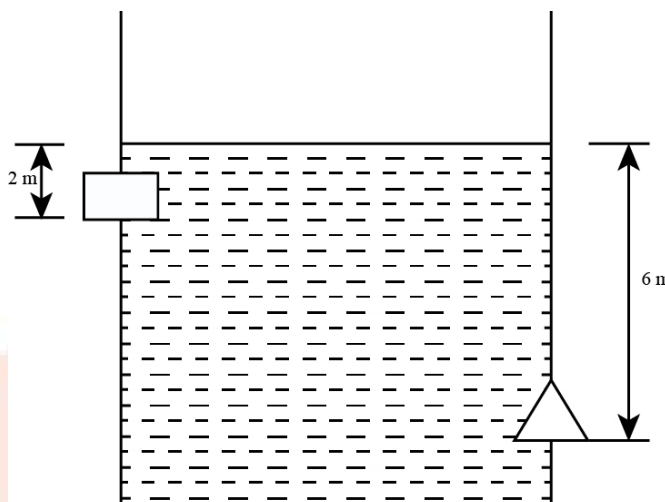
$$\frac{\Delta L_s}{\Delta L_b} = \left(\frac{F_s}{F_b} \right) \left(\frac{L_s}{L_b} \right) \left(\frac{A_b}{A_s} \right) \left(\frac{E_b}{E_s} \right)$$

$$\frac{\Delta L_s}{\Delta L_b} = \frac{7}{4}(a) \left(\frac{1}{b} \right) \left(\frac{1}{c} \right)$$

$$\frac{\Delta L_s}{\Delta L_b} = \frac{7a}{4bc}$$

$$\frac{\Delta L_b}{\Delta L_s} = \frac{4bc}{7a}$$

94. The figure below represents the tanks with the two holes,



The formula for the velocity is,

$$v = \sqrt{2gh}$$

The volumetric flow rate of square hole is,

$$\begin{aligned} V_s &= Av \\ &= x^2 \times \sqrt{2g \times 2} \end{aligned}$$

The volumetric flow rate of triangular hole is,

$$\begin{aligned} V_t &= Av \\ &= \frac{\sqrt{3}}{4} \times 4^2 \times \sqrt{2g \times 6} \end{aligned}$$

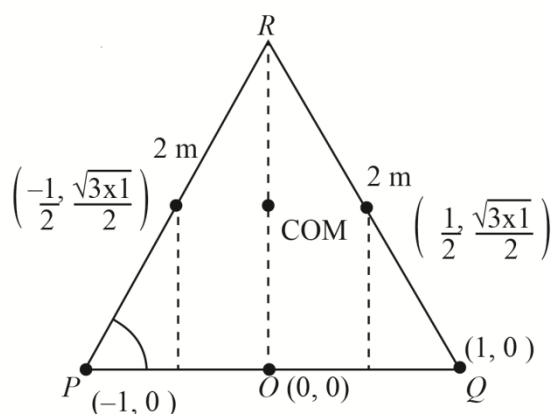
Since the flow rate is same,

$$\begin{aligned} V_s &= V_t \\ x^2 \times \sqrt{2g \times 2} &= \frac{\sqrt{3}}{4} \times 4^2 \times \sqrt{2g \times 6} \end{aligned}$$

$$x^2 = \sqrt{3} \times \sqrt{3}$$

$$x = \sqrt{3} = 1.732 \text{ cm}$$

95. The figure below represents the three uniform thin rods in the form of triangle.



The centre of mass of the system,

$$r_{COM} = \frac{m_1 r_1 + m_2 r_2 + m_3 r_3}{(m_1 + m_2 + m_3)}$$

$$(r_{COM})_i = \frac{1}{2} \left(\frac{\sqrt{3}}{2} \times 1 \times 2 \right) \hat{j}$$

$$(r_{COM})_i = \frac{1}{\sqrt{3}} \hat{j}$$

The new length of the rod due to expansion is,

$$l_f = l_i (1 + \alpha \Delta T)$$

$$l_f = l_i \left(1 + \frac{\gamma}{3} \Delta T \right)$$

Substitute the values in the above equation.

$$l_f = 2 \left(1 + \frac{12\sqrt{3} \times 10^{-6}}{3} \times 50 \right)$$

$$l_f = 2 \left(1 + 4 \times 50 \times \sqrt{3} \times 10^{-6} \right)$$

$$l_f = 2 \left(1 + 2\sqrt{3} \times 10^{-4} \right)$$

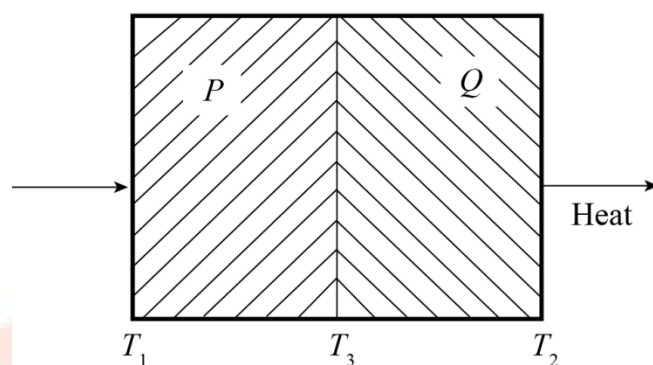
The final centre of mass can be calculated as,

$$\begin{aligned} (r_{COM})_f &= \frac{1}{3} \left(\frac{\sqrt{3}}{2} \times (1 + 2\sqrt{3} \times 10^{-4}) \times 2 \right) \hat{j} \\ &= \left(\frac{1}{\sqrt{3}} + 2 \times 10^{-4} \right) \hat{j} \end{aligned}$$

The increase in y coordinate of the centre of mass of the system of the rods is,

$$\begin{aligned} \delta y &= (r_{COM})_f - (r_{COM})_i \\ &= \left(\frac{1}{\sqrt{3}} + 2 \times 10^{-4} - \frac{1}{\sqrt{3}} \right) \hat{j} \\ &= 2 \times 10^{-4} \text{ m} \end{aligned}$$

96. The figure below represents the wall P and Q.



It is given that,

$$k_Q = \frac{k_P}{2}$$

$$k_P = 2k_Q$$

And

$$T_1 - T_2 = 24^\circ\text{C}$$

In steady state, the rate of flow heat through layer P and Q is same,

$$\frac{k_P a (T_1 - T_3)}{x} = \frac{k_Q a (T_3 - T_2)}{x}$$

Substitute the values in the above equation.

$$\frac{2k_Q a (T_1 - T_3)}{x} = \frac{k_Q a (T_3 - T_2)}{x}$$

$$2(T_1 - T_3) = (T_3 - T_2)$$

$$2T_1 - 3T_3 + T_2 = 0$$

From equation (I),

$$2T_1 - 3T_3 + T_1 - 24 = 0$$

$$3T_1 - 3T_0 = 24$$

$$T_1 - T_0 = 8^\circ\text{C}$$

97. The relation is given as,

$$V = KT^2$$

From ideal gas equation,

$$PV = RT$$

$$PV = R\left(\frac{V}{K}\right)^{1/2}$$

$$PV^{1/2} = a \text{ const}$$

From the above equation, $\gamma = 1/2$.

The work done by the gas is,

$$W = \frac{nR(T_i - T_f)}{1 - \gamma}$$

$$W = \frac{R(60)}{1 - 1/2}$$

$$W = 120R$$

98. The total work done is equal to the area under the cyclic process.

$$W = (3P - P)(2V - V)$$

$$W = 2PV$$

The total heat supplied to the system,

$$\begin{aligned}
 Q &= Q_{AB} + Q_{BC} \\
 &= nc_v \Delta T + nc_p \Delta T \\
 &= \frac{3}{2} R \Delta T + \frac{5}{2} R \Delta T \\
 &= \frac{3}{2} V \Delta P + \frac{5}{2} 3P \Delta V \\
 &= \frac{3}{2} V (3P - 2P) + \frac{5}{2} 3P (2V - V) \\
 &= 3PV + \frac{15}{2} PV \\
 &= \frac{21}{2} PV
 \end{aligned}$$

The efficiency is calculated as,

$$\begin{aligned}
 \eta &= \frac{W}{Q} \times 100 \\
 &= \frac{2PV}{\frac{21}{2} PV} \times 100 \\
 &= 19.04\%
 \end{aligned}$$

Thus, no option is correct.

99. The translation kinetic energy of a molecule of gas is $= \frac{3}{2} kT$

The formula for the kinetic energy is,

$$\frac{3}{2}kT = eV$$

$$\frac{3}{2}(1.38 \times 10^{-23})T = (1.6 \times 10^{-19})(10)$$

$$T = 7.73 \times 10^4$$

$$T = 77.3 \times 10^3 \text{ K}$$

100. The equation of the transverse displacement is,

$$y = 0.03 \sin\left(\frac{2\pi x}{3}\right) \cos 60\pi t \quad \dots\dots (I)$$

The mass per unit length is given as, $\mu = 0.01 \text{ kg/m}$

The general equation of transverse displacement is,

$$y = a \sin kx \cos \omega t$$

Comparing the above equation with equation (I),

$$\omega = 60\pi, k = \frac{2\pi}{3}$$

The formula for the velocity is,

$$v = \sqrt{\frac{T}{\mu}} = \frac{\omega}{k}$$

Substitute the values in the above equation.

$$\sqrt{\frac{T}{0.01}} = \frac{60\pi}{2\pi/3}$$

$$\sqrt{\frac{T}{0.01}} = 90$$

$$T = 90^2 \times 0.01$$

$$T = 81 \text{ N}$$

101. For the closed organ pipe,

$$f = f_2 = 3f_1$$

Now,

$$3f_1 = \frac{3v}{4L} = \frac{3}{4L} \sqrt{\frac{\gamma p}{\rho_1}}$$

For the open organ pipe,

$$f = f_2 = 2f_1$$

Now,

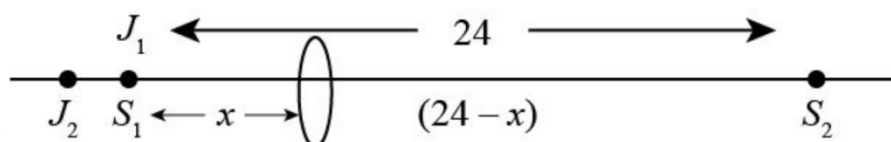
$$2f_1 = \frac{2v}{2L'} = \frac{1}{L'} \sqrt{\frac{\gamma p}{\rho_2}}$$

From same sound frequency,

$$\frac{3}{4L} \sqrt{\frac{\gamma p}{\rho_1}} = \frac{1}{L'} \sqrt{\frac{\gamma p}{\rho_2}}$$

$$L' = \frac{4L}{3} \sqrt{\frac{\rho_1}{\rho_2}}$$

102. The figure below represents the convex lens and two point sources,



The given condition satisfies when S_1 is placed in one side such that $u < f$ and the S_2 is placed on the other side of the lens such that $u > f$.

S_1 is the object for lens, from lens formula,

$$\frac{1}{f} = \frac{1}{-y} - \frac{1}{-x}$$

$$\frac{1}{y} = \frac{1}{x} - \frac{1}{-f} \quad \text{..... (I)}$$

S_2 is the object for the lens, then

$$\frac{1}{f} = \frac{1}{y} - \frac{1}{-(24-x)}$$

$$\frac{1}{-y} = \frac{1}{f} - \frac{1}{24-x} \quad \text{..... (II)}$$

From equation (I) and (II),

$$\frac{1}{x} - \frac{1}{f} = \frac{1}{f} - \frac{1}{24-x}$$

$$\frac{1}{x} - \frac{1}{9} = \frac{1}{9} - \frac{1}{24-x}$$

$$\frac{1}{x} + \frac{1}{24-x} = \frac{2}{9}$$

$$x^2 - 24x + 108 = 0$$

By solving the above equation,

$$x = 6 \text{ cm or } 18 \text{ cm}$$

103. The formula for the fringe width is,

$$\beta = \frac{\lambda d}{D}$$

Substitute the given values in the above equation.

$$\beta = \frac{(120 \times 10^{-2})(0.5 \times 10^{-9})}{500 \times 10^{-9}}$$

$$\beta = 12 \times 10^{-4} \text{ m}$$

The number of fringes in 3 mm distance on screen is,

$$n = \frac{3 \text{ mm}}{12 \times 10^{-4}}$$

$$n = \frac{3 \times 10^{-3}}{12 \times 10^{-4}} = 25$$

That is third fringe.

The phase difference is,

$$\phi = 2\pi + 2\pi + \pi$$

$$\phi = 5\pi$$

104. The acceleration of projected charge in y-direction is,

$$a_y = -\frac{F}{m}$$

$$a_y = -\frac{qE}{m}$$

Substitute the given values in the above equation.

$$a_y = -\frac{(1 \times 10^{-6})(1 \times 10^3)}{2 \times 10^{-3}}$$

$$a_y = -\frac{1}{2} \text{ m/s}^2$$

The vertical component of the projection velocity is,

$$u_y = u \sin 45^\circ$$

If the particle is not to hit upper plate, then

$$(u_y)^2 \leq 2a_y(y)$$

$$(u \sin 45^\circ)^2 \leq 2 \times \frac{1}{2} \times 2 \times 10^{-2}$$

$$u^2 \leq 4 \times 10^{-2}$$

$$u \leq 2 \times 10^{-1}$$

$$u \leq 0.2 \text{ m/s}$$

105. The formula for the intensity of the electric field due to infinitely long wire at distance r is,

$$E = \frac{\lambda}{2\pi\epsilon_0 r}$$

$$E = \frac{2\lambda}{4\pi\epsilon_0 r}$$

Substitute the given values in the above equation.

$$E = \frac{(9 \times 10^9) 2 \left(\frac{1}{3}\right)}{3 \times 18 \times 10^{-2}}$$

$$E = \frac{1}{3} \times 10^{11} \text{ N/C}$$

The force on the charge is,

$$F = qE$$

$$= 3 \times 10^{-6} \times \frac{1}{3} \times 10^{11}$$

$$= 10^5 \text{ N}$$

106. The voltage across capacitance is,

$$\begin{aligned}V_{cap} &= \frac{Q}{C} \\&= \frac{4 \times 10^{-6}}{2 \times 10^{-6}} \\&= 2 \text{ V}\end{aligned}$$

The across 2Ω resistance is same as of V_{Cap}

Therefore,

$$\begin{aligned}I &= \frac{V}{R} \\I &= \frac{2}{2} = 1 \text{ A}\end{aligned}$$

The formula for the emf of the cell,

$$E = V + Ir$$

Substitute the values in the above equation.

$$\begin{aligned}E &= 2 + 1 \times 0.5 \\&= 2.5 \text{ V}\end{aligned}$$

107. The diameter of the sphere is, $d = 18 \text{ cm} = 18 \times 10^{-2} \text{ m}$

Potential: $V = 25 \text{ kV} = 25 \times 10^3 \text{ V}$

The formula for the charge on the metal sphere is,

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r} \right)$$

$$q = \frac{4\pi\epsilon_0(r)}{q}$$

Substitute the values in the above equation.

$$q = \frac{25 \times 10^3 \times \frac{18}{2} \times 10^{-2}}{9 \times 10^9}$$

$$q = 0.25 \times 10^{-6}$$

$$q = 0.25 \mu\text{C}$$

108. Given data:

$$V = 1.6 \text{ V}$$

$$R = 4 \Omega$$

The current in the wire is,

$$I_1 = \frac{1.6}{4} = 0.4 \text{ A}$$

The emf of the cell is,

$$E = 1.6 + 0.4r \quad \text{..... (I)}$$

When a 4Ω resistance is connected in parallel,

$$\frac{1}{R'} = \frac{1}{4} + \frac{1}{4}$$

$$\frac{1}{R'} = \frac{2}{4}$$

$$R' = 2 \Omega$$

And

$$I_2 = \frac{1.33}{2} = 0.66 \text{ A}$$

$$E = 1.33 + 0.66r \quad \text{..... (II)}$$

From equation (I) and (II),

$$1.6 + 0.4r = 1.33 + 0.66r$$

$$1.6 - 1.33 = 0.66r - 0.4r$$

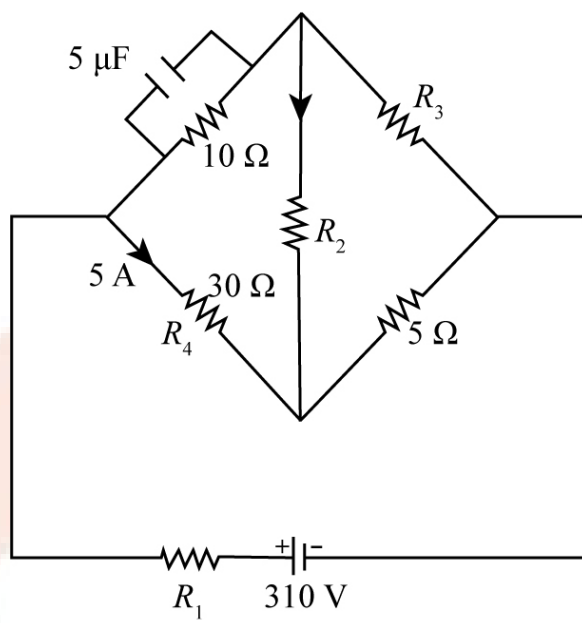
$$r = \frac{0.27}{0.26} \approx 1$$

Substitute the value in equation (I).

$$E = 1.6 + 0.4 \times 1$$

$$E = 2 \text{ V}$$

109. The figure below represents the given circuit.



$$V_{cap} = \frac{Q}{C}$$

$$V_{cap} = \frac{1 \times 10^{-3}}{5 \times 10^{-6}}$$

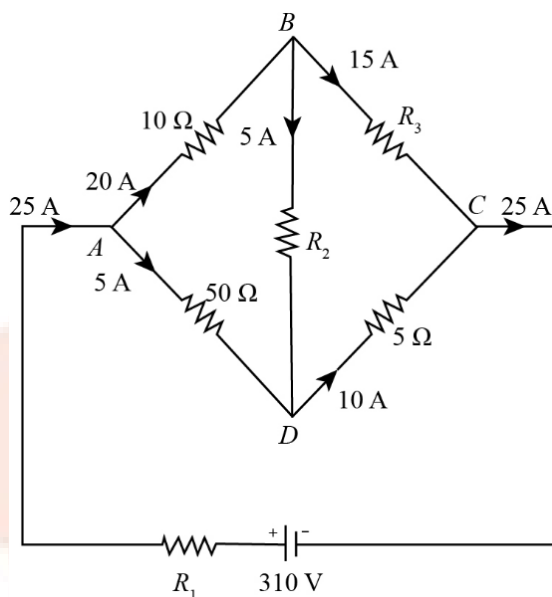
$$V_{cap} = 200\ \text{V}$$

The current across R_5 resistance,

$$I = \frac{V}{R}$$

$$I = \frac{200}{10} = 20\ \text{A}$$

The figure below represents the current distribution.



In loop $BCDB$,

$$-15R_3 + 10 \times 5 + 5R_2 = 0 \quad \text{..... (I)}$$

In loop $ABCD$,

$$-200 - 15R_3 + 50 + 250 = 0$$

$$15R_3 = 100$$

$$R_3 = \frac{100}{15} \Omega$$

From equation (I),

$$5R_2 = 50$$

$$R_2 = 10 \Omega$$

In loop $ADCA$,

$$-250 - 50 + 310 - 25R_1 = 0$$

$$10 = 25R_1$$

$$R_1 = \frac{2}{5} \Omega$$

The required value is,

$$\frac{R_1 \times R_2}{R_3} = \frac{\frac{2}{3} \times 10}{\left(\frac{100}{15}\right)}$$

$$= \frac{3}{5} = 0.6$$

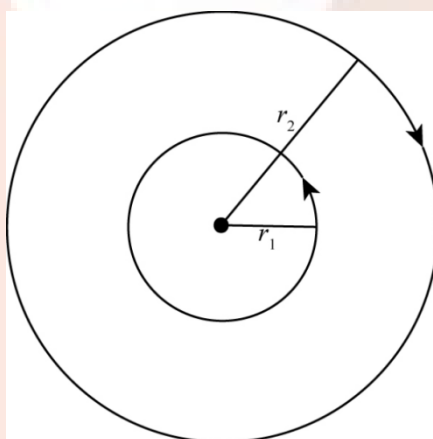
110. Given Data:

$$I_1 = 0.4 \text{ A}, I_2 = 0.6 \text{ A},$$

$$r_1 = 30 \times 10^{-2} \text{ m}$$

$$r_2 = 60 \times 10^{-2} \text{ m}$$

Consider the figure,



The expression for the magnetic field at O,

$$B_O = \frac{\mu_0}{4\pi} \cdot 2\pi \left[\frac{n_1 I_1}{r_1} - \frac{n_2 I_2}{r_2} \right]$$

Substitute the values in the above equation.

$$B_O = \frac{\mu_0}{4\pi} \cdot 2\pi \left[\frac{20 \times 0.4}{30 \times 10^{-2}} - \frac{20 \times 0.6}{60 \times 10^{-2}} \right]$$

Further simplification,

$$B_O = \frac{10}{3} \mu_0$$

111. The expression for the force per unit length is,

$$\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi x}$$

The expression for the work done per unit length is,

$$W = F \cdot dx$$

$$W = \frac{\mu_0 I_1 I_2}{2\pi x} \cdot dx$$

$$W = \frac{\mu_0 I_1 I_2}{2\pi} \int \frac{dx}{x}$$

$$W = \frac{\mu_0 I_1 I_2}{2\pi} \log_e x$$

$$W \propto \log_e x$$

112. The correct matching is shown below,

(a)	High Retentivity	(iv)	Permanent magnet
(b)	High resistivity	(iii)	To decrease eddy current losses

(c)	Low coercivity	(i)	Telephone diaphragm
(d)	Negative susceptibility	(ii)	Diamagnet

Thus correct option is (2).

113. Due to the repulsion caused by emf induced, it is more difficult to push a magnet into a coil with a number of turns

The emf induced in the coil always opposes the motion of a magnet.

114. The power is expressed as,

$$P = I^2 R$$

Substitute the values in the given equation.

$$240 = 4^2 R$$

$$R = 15 \Omega$$

For L - R circuit,

$$V = IZ$$

$$V = I\sqrt{X_L^2 + R^2}$$

$$\frac{V^2}{I^2} = X_L^2 + R^2$$

Substitute the values in the above equation.

$$\frac{100^2}{4^2} = X_L^2 + 15^2$$

$$X_L^2 = 625 - 225$$

$$X_L^2 = 400$$

$$X_L = 20$$

And

$$X_L = L\omega$$

$$X_L = L(2\pi f)$$

$$20 = L(2\pi \times 50)$$

$$L = \frac{1}{5\pi} \text{ H}$$

115. The expression for the speed of the electromagnetic wave is,

$$c = \frac{E_0}{B_0}$$

$$3 \times 10^8 = \frac{60}{B_0}$$

$$B_0 = 2 \times 10^{-7} \text{ T}$$

116. From the de-Broglie wavelength of electron, the kinetic energy of the electron,

$$\lambda_e = \frac{h}{\sqrt{mK_e}}$$

$$K_e = \frac{h^2}{\lambda^2 \times 2m_e}$$

The kinetic energy of photon is,

$$K_p = h\nu = \frac{hc}{\lambda_p}$$

The required ratio is,

$$\frac{K_e}{K_p} = \frac{h^2 / \lambda^2 2m_e}{hc / \lambda}$$

$$\frac{K_e}{K_p} = \frac{h}{c\lambda \times 2m_e}$$

Substitute the given values in the above equation.

$$\frac{K_e}{K_p} = \frac{6.66 \times 10^{-34}}{3 \times 10^8 \times 1.2 \times 10^{-10} \times 2 \times 9.1 \times 10^{-31}}$$

$$\frac{K_e}{K_p} \approx \frac{1}{100}$$

117. Given Data,

$$r_1 = 5.5 \times 10^{-11} \text{ m}$$

$$v_1 = 4 \times 10^6 \text{ m/s}$$

The expression for the first excited state is,

$$r_n \propto \frac{n^2}{Z}$$

$$\frac{r_n}{2^2} = \frac{5.5 \times 10^{-11}}{1^2}$$

$$r_n = 22 \times 10^{-11} \text{ m/s}$$

The speed of the electron,

$$v_n \propto \frac{Z}{n}$$

$$v_n = \frac{4 \times 10^6}{2} = 2 \times 10^6 \text{ m/s}$$

The time period of revolution,

$$\begin{aligned} T &= \frac{2\pi r}{v_n} \\ &= \frac{2 \times 3.14 \times 22 \times 10^{-11}}{2 \times 10^6} \\ &= 6.908 \times 10^{-16} \text{ s} \end{aligned}$$

118. From the kinetic energy formula,

$$v^2 = \frac{2 \times KE}{m}$$

Substitute the given values in the above equation.

$$v^2 = \frac{2 \times 4 \times 1.6 \times 10^{-19}}{3.2 \times 10^{-21}}$$

$$v^2 = 400$$

$$v = 20 \text{ m/s}$$

The time required to travel a distance of 3.6 km is,

$$t = \frac{D}{v}$$

$$t = \frac{3.6 \times 10^3}{20}$$

$$= 180 \text{ s or } 3 \text{ min}$$

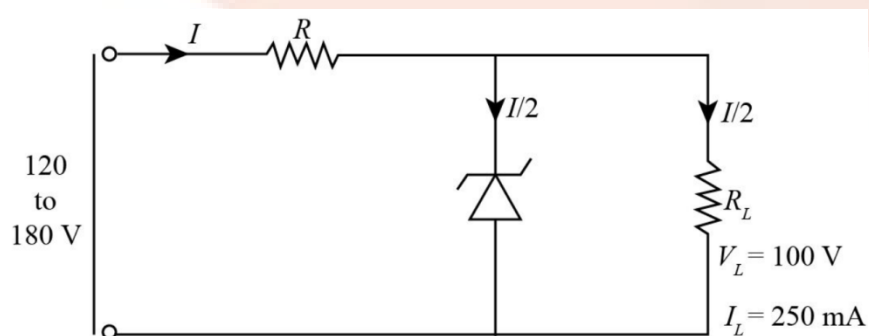
The number of particle decays in 3 min,

$$n = \frac{N_0 - N}{N_0} \times 100$$

$$n = \frac{N_0 - N_0 / 8}{N_0} \times 100$$

$$n = \frac{7}{8} \times 100 = 87.5\%$$

119. The figure below represents the electric circuit with zener diode.



$$\begin{aligned} R_L &= \frac{V_L}{I_L} \\ &= \frac{110}{250 \times 10^{-3}} = 440 \, \Omega \end{aligned}$$

The current will equally shared by zener diode and resistance.

$$I = 2 \times 250 = 500 \, \text{mA}$$

Therefore,

$$\begin{aligned} R_S &= \frac{V_S}{I} \\ &= \frac{180 - 110}{0.5} \\ &= 140 \, \Omega \end{aligned}$$

120. The process of recovering the modulated signal from the modulated carrier wave is called demodulation.

